

Tutorial on Ant Colony Optimization

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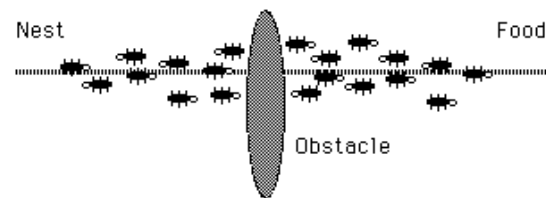
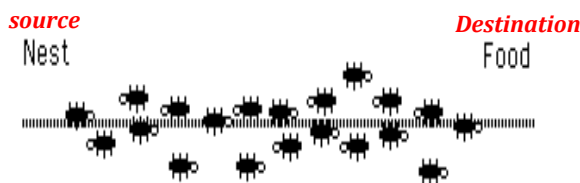
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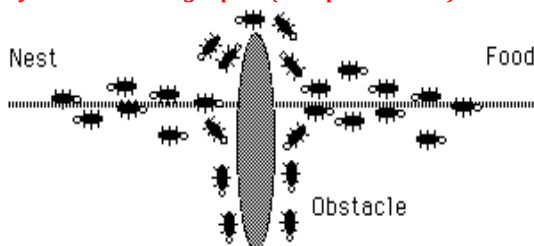
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Ant Colony Optimization (ACO) studies artificial systems that take inspiration from the *behavior of real ant colonies* and which are used to solve discrete optimization problems.” First introduced by Marco Dorigo in 1992. Originally applied to Traveling Salesman Problem. Natural behavior of ants have inspired scientists to mimic insect operational methods to solve real-life complex optimization problems. By observing ant behavior, scientists have begun to understand their means of communication. Ant-based behavioral patterns to address combinatorial problems - first proposed by Marco Dorigo.

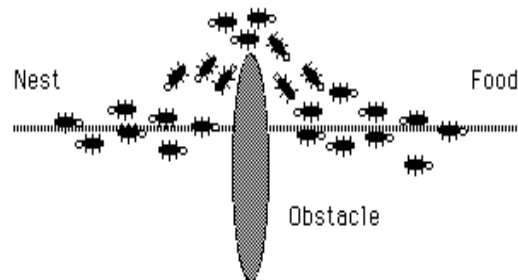
Ants secrete *pheromone* while traveling from the nest to food, and vice versa in order to communicate with one another to find the shortest path



Now, Two Paths Exist from source (nest) to destination (food). initial: 50% - 50% split ...equal ants in both paths
But SHORTER path deposit more pheromone DENSITY (why? explain by example: 100 + 100 ants (10m + 20 m distance => x times vs. 2x times pass => double PH. density on the HALF length path (10m path vs. 20m))



Over the time, All ants Gradually converge to this single path that is shorter (because, higher pheromone density)



Ants are forced to decide whether they should go left or right, and the choice that is made is a random decision. **Pheromone accumulation is faster on the shorter path. The difference in pheromone content between the two paths over time makes the ants choose the shorter path.**

The more ants follow a trail, the more attractive that trail becomes for being followed.

Different optimization problems have been explored using a simulation of this real ant behavior

In ACO, the search space is represented as a Graph. For example, In TSP => each city is represented by a Vertex (or, Node) of the graph. Each edge between pair (r, s) of cities has a 'pheromone' value and a 'visibility' values, whose priority denoted by the TAO and ETA functions, respectively (see following Eq. 8.1)

An ant k at node r will choose the destination node s at a later stage with probability

$$p_k(r, s) = \begin{cases} \tau(r, s)^\alpha \eta(r, s)^\beta / \sum_{u \in M_k} \tau(r, u)^\alpha \eta(r, u)^\beta, & \text{for } s \in M_k \\ 0, & \text{otherwise} \end{cases} \quad (8.1)$$

where α indicates the degree of importance of pheromone, β -degree visibility and $u \in M_k$ is a choice that belongs ant k (neighborhood) when he was at node r. Neighborhood of ant k at node r will contain all the nodes that can be addressed directly connected to the node r, except nodes that previously already visited. Visibility usually be replaced with $1 /$ (distance between node r to node s)

$p_k(r, s)$: Probability that Ant 'k' chooses the city 's' as the next city from its current city 'r' (i.e., Ant k is currently at city r)

$\tau(r, s)$: Pheromone density between city r and city s Alpha: degree of importance of Pheromone

$\eta(r, s)$: Visibility between city r and city s = $1.0 /$ distance between city r and s Beta: degree of importance of Visibility

M_k : set of allowed cities as NEXT city for Ant k (i.e., Cities where Ant k can go NEXT, discarding previously visited & current city)

An ant k when passing a segment will leave pheromone . The amount of pheromone contained in the segment ij after passed by ant k is given by the formula

After Ant k moves from one city to another, and thus constructs the whole Tour (cycle), it then Deposits pheromone $\Delta \tau_{i,j}^k$ on All the edges (i, j) Along its Tour.

$$\begin{array}{l} \text{Amount of Pheromone} \leftarrow \tau_{i,j} \leftarrow \tau_{i,j} + \Delta \tau_{i,j}^{(k)} \\ \text{between city i and j} \end{array} \quad \begin{array}{l} \text{New Pheromone deposited by Ant k on edge (i,j)} \\ \text{if the edge (i.e., city-pair (i, j)) belong to its Tour (path)} \end{array} \quad (8.2)$$

With the increasing value of pheromone on segment i-j, then the probability of this segment to be chosen by ants at the next iteration increases. Once a node is passed then the pheromone evaporation will occur with the following rules

Eq. 8.3 simulates Evaporation of Pheromone on Edge (i,j) i.e., between city i and city j

$$\tau_{i,j} \leftarrow (1 - \rho) \tau_{i,j} ; \forall (i, j) \in A \quad (8.3)$$

Example: $\rho = 0.2$ means 20% evaporation of Pheromone (i.e., 80% of Pheromone remains during next iteration)

where $\rho \in (0, 1]$ is the evaporation rate parameter and A represents segments that have been passed by ant k as part of the path from the nest to get to the food. The decline in the amount of pheromone allows ants to explore a different path during the search process. It also would eliminate the possibility of choosing the path that is not good enough. The amount of pheromone is added to the segment i - j by ant k is given as

$$\Delta \tau_{i,j}^{(k)} = \frac{Q}{L_k} \quad (8.4)$$

where Q is a constant and L_k is the total distance traveled by ant k after returning to the nest (in the case of the TSP is the path traversed by the salesman). Q value is usually

↓
Length of the

This ensures Good TSP Tours (with Small value of Length L) can get MORE Pheromone than Worse TSP Tours (because L is larger in Longer worse Tours)

determined by the user. In many cases the value of $Q = 1$ gives good results. Or it could also be implemented in the following way

This means Only the Edges Along the Best Tour get this Pheromone Boost !

c: Multiplier for Pheromone Boost

(Large c => Large Boost)

fbest: Fitness of Best Path (1/Length)

fworst: Fitness (i.e., 1.0/Length) of worst Path

$$\Delta\tau_{i,j}^{(k)} = \begin{cases} \frac{cf_{best}}{f_{worst}}, & \text{if } (i,j) \in \text{best global path} \\ 0, & \text{otherwise} \end{cases} \quad (8.5)$$

where $fbest$ is the best value of the objective function and $fworst$ is the worst value of the objective function. While c is a constant to control the scale of the global pheromone updating. The higher value of c the more pheromone is added to the global path of the best and the better ability to exploit. Equation (8.5) is intended to provide more amount of pheromone on the path of global best (overall).

ACO Algorithm for TSP

Repeat

- Randomly place ants at the cities
- For each ant:
 1. ^{Probabilistically} Choose a not yet visited city ^{using following equation (write the Eq. 8.1)} until a tour is completed
 2. optimize the tour *by mutations / heuristics [However, this is NOT done in the following example]*
 3. Update pheromone $\tau_{ij} += 1/\text{length}(\text{tour})$ *that is, Eq. 8.2 and 8.4*
- Evaporate Pheromone $\tau_{ij} = (1 - \rho) \cdot \tau_{ij}$ *i.e., Eq. 8.3*

Until the fixed run time / no. of iterations is exceeded

Example

Suppose we have a TSP problem with 5 cities. The distance between cities is given by the following matrix

d =					
	0	10	12	11	14
	10	0	13	15	8
	12	13	0	9	14
	11	15	9	0	16
	14	8	14	16	0

This is the Pairwise Distance between Every Pair of Cities

d_{ij} = Distance between city i and city j = d_{ji} [thus, d is symmetric matrix]

For example, d₁₃ = d₃₁ = 12, which is distance between city 1 and city 3

Fig. 2: Distance Matrix

Determine the best route with minimum total distance by using the ACO.

Thus, we have 3 Ants. For simplicity of demonstration, we assume that All 3 Ants are initially Placed at city 1. Also, initial $\tau_{00} = 1$ for all city pairs, $\alpha = 1$, $\beta = 2$, next two Random numbers in pipeline are 0.6841 and 0.4024. Construct the Tour upto the 2nd iteration using this data

Step 1:

Suppose the number of ants, $N = 3$. We determine the visibility (h) between the cities by taking the inverse of the distance, $1 / d$. we get

$h =$

0	0.1000	0.0833	0.0909	0.0714
0.1000	0	0.0769	0.0667	0.1250
0.0833	0.0769	0	0.1111	0.0714
0.0909	0.0667	0.1111	0	0.0625
0.0714	0.1250	0.0714	0.0625	0

visibility = $1/\text{distance}$

Fig. 3: Visibility Matrix $h = 1/d$

Then for each segment between cities we give the initial pheromone value τ_0 , suppose this value is the same for all segments, namely 1.

$\tau_{\text{init}} =$

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

τ_{ij} = Pheromone between city i and city j = initially, all 1

Fig. 4: Initial Pheromone Matrix τ_0

Step 2:

Determine the probability of moving from one city to another. In this case city 1 is designated as a departure city. Then from the last city visited we are going back to city 1.

As city 1 is chosen as the beginning city 1 then city will be taboo to visit again, so that the level of visibility of the city 1 is made =0.

$h =$

0	0.1000	0.0833	0.0909	0.0714
0	0	0.0769	0.0667	0.1250
0	0.0769	0	0.1111	0.0714
0	0.0667	0.1111	0	0.0625
0	0.1250	0.0714	0.0625	0

Fig. 5: Visibility Matrix when partial Tour = 1 - ...

Then we calculate the possibility to visit other cities from city 1 using the formula in equation (8.9) .

$$p_k(r, s) = \begin{cases} \tau(r, s)^\alpha \eta(r, s)^\beta / \sum_{u \in M_k} \tau(r, u)^\alpha \eta(r, u)^\beta, & \text{if } s \in M_k \\ 0, & \text{otherwise} \end{cases} \quad (8.9)$$

Where $p_k(r, s)$ is probability move from city r to city s of ant k , τ is pheromone level, $\eta(r, s)$ is visibility of cities r and s , M_k set of cities possibly visited by ant k , α weight of pheromone τ and β is weight to control visibility. Using $\alpha=1$ and $\beta=2$.

Suppose ants 1 from city 1 will choose the next city, then we calculate

$\tau(1, s)^1 \eta(1, s)^2$ from 1 city to all towns *(Set of allowed cities $M_k = \{2, 3, 4, 5\}$. Thus, Put $s = 2, 3, 4, 5$ one by one)*

$1 \times (0.1000)^2 = 0.01$ (from city 1 to city 2)

$1 \times (0.0833)^2 = 0.0069$ (from city 1 to city 3)

$1 \times (0.0909)^2 = 0.0083$ (from city 1 to city 4)

$1 \times (0.0714)^2 = 0.0051$ (from city 1 to city 5)

Sum of these above FOUR values = $0.01 + 0.0069 + 0.0083 + 0.0051 = 0.0303$

The total number of $\tau(1, s)^1 \eta(1, s)^2 = 0.0303$

h =					
0	0.1000	0.0833	0.0909	0.0714	
0	0	0.0769	0.0667	0.1250	
0	0.0769	0	0.1111	0.0714	
0	0.0667	0.1111	0	0.0625	
0	0.1250	0.0714	0.0625	0	

So probability of ant 1 going from city 1 to all cities respectively calculated by the formula

$$p_1(1, s) = \tau(1, s)^1 \eta(1, s)^2 / \sum \tau(1, s)^1 \eta(1, s)^2$$

Obtained the probability to go from

Probability

of City 1 to 2 = $0.01 / 0.0303 = 0.3299$

City 1 to 3 = $0.0069 / 0.0303 = 0.2291$

City 1 to 4 = $0.0083 / 0.0303 = 0.2727$

City 1 to 5 = $0.0051 / 0.0303 = 0.1683$

The cumulative number of these probabilities is

Cumulative Probability

Upto City 2 = 0.3299

Upto City 3 = 0.5590

Upto City 4 = 0.8317

Upto City 5 = 1

This is required to implement a selection scheme similar to the Roulette Wheel Selection (RWS) which is actually a Fitness Proportional Selection strategy

Step 3:

We generate a random number r to determine the choice of cities to be visited by comparing it with the value of the cumulative probability. Suppose the generate random numbers $r = 0.6841$, then we can compare it with the cumulative value of probability. This number is located below the 0.8317 on the cumulative amount. Thus the city to visit is city

city 4. *Thus, Tour constructed so far by ANT 1 is: 1-4- ...*

Since city 4 has been selected then 4th column of the matrix, the visibility set to 0 so that

$h =$

0	0.1000	0.0833	0	0.0714
0	0	0.0769	0	0.1250
0	0.0769	0	0	0.0714
0	0.0667	0.1111	0	0.0625
0	0.1250	0.0714	0	0

*This means, city 1 and 4 are now NOT allowed in the Tour as they are already visited !
See: $h_{14} = h_{24} = h_{34} = h_{44} = h_{54} = 0$, and $h_{11} = h_{21} = h_{31} = h_{41} = h_{51} = 0$
thus making i -th column 0 makes i -th city INVISIBLE from ALL other cities, so the i -th city WONT be Repeated in the tour ever again :)*

From here we calculate again the value of τ (Now, from city 4, except city 1. Thus $M_k = \{2, 3, 5\}$)
 $\tau(4, s)^1 \eta(4, s)^2$ from city 4 to all of the cities (except city 1)

$$1 \times (0.0667)^2 = 0.0044 \text{ (from city 4 to city 2)}$$

$$1 \times (0.1111)^2 = 0.0123 \text{ (from city 4 to city 3)}$$

$$1 \times (0)^2 = 0.0 \text{ (from city 4 to city 4 or city 1)}$$

$$1 \times (0.0625)^2 = 0.0039 \text{ (from city 4 to city 5)}$$

$$\begin{aligned} \text{Sum of these values} &= \\ &0.0044 + 0.0123 + 0.0039 \\ &= 0.0207 \end{aligned}$$

The total number of $\tau(1,s)^1\eta(1,s)^2 = 0.0207$

So probability ant 1 going from city 4 to other cities respectively calculated by the formula

$$p_1(4,s) = \tau(4,s)^1\eta(4,s)^2 / \sum \tau(4,s)^1\eta(4,s)^2$$

Obtained the probability to go from

Probability

of City 4 to 2 = $0.0044 / 0.0207 = 0.2147$

City 4 to 3 = $0.0123 / 0.0207 = 0.5965$

City 4 to 4 = $0.0 / 0.0207 = 0.0$

City 4 to 5 = $0.0039 / 0.0207 = 0.1887$

Suppose we generate a random number $r = 0.4024$.

Total cumulative

City 2 = 0.2147

City 3 = 0.8113

City 5 = 1.0

Because the random numbers $r = 0.4024 < 0.8113$, then the next city selected is city 3. Thus the route formed for ant 1 so far is 1-4- 3.

This step is repeated for all of ants. Of each route passed through by every ant, we will know the total distance traversed by each ant. Suppose the calculation we did in the first iteration for all 3 ants we obtained the following

Ants 1: 1-4 - 3 - 5-2 -1 **$L = 11 + 9 + 14 + 8 + 10 = 52$**

Ants 2: 1 - 4-2 - 5 - 3-1 **$L = 11+15+8+14+12 = 60$**

Ants 3: 1 - 4-5 - 2 - 3-1 **$L = 11+16+8+13+12 = 60$**

The most common edges =

1-4 (3 times), 5-2 (2 times), 3-1 (2 times)

They will receive Maximum Pheromones

edge 1-4 will get pherom. from All 3 Ants (highest)

edge 5-2 and 3-1 will get pherom. from 2 Ants

*** See Fig. 6 for Pheromone Values ***

With a total distance of each route is 52, 60 and 60. The total distance is used to update the pheromone level τ using the formula (8:10).

$$\tau_{r,s} \leftarrow (1 - \rho)\tau_{r,s} + \sum_{k=1}^N \Delta\tau_{r,s}^k \quad (8.10)$$

Step 4:

d =					
	0	10	12	11	14
10		0	13	15	8
12	13		0	9	14
11	15	9		0	16
14	8	14	16		0

All ants will provide additional pheromone to the pheromone matrix according to the route passed through.

Ants will provide additional pheromone by $1/\text{total distance}$ i.e. $1/52 = 0.0192$. In addition, evaporation will occur at $(1-0.5) \times 1 = 0.5$, where ρ , evaporation coefficient = 0.5. So the total pheromone becomes **0.5192**. In other segments which are not passed by ants, only evaporation will occur without the addition of pheromone.

These changes will be given to the segments passed through by ants 1 at the pheromone matrix such that

0.5000	0.5000	0.5000	0.5192	0.5000
0.5192	0.5000	0.5000	0.5000	0.5000
0.5000	0.5000	0.5000	0.5000	0.5192
0.5000	0.5000	0.5192	0.5000	0.5000
0.5000	0.5192	0.5000	0.5000	0.5000

Fig. 6: Updated Pheromone Matrix by Ant 1

Evaporation: All Edges (row = 0.5)
All Pheromones $1.0 \Rightarrow 0.5$

Tour = 1-4-3-5-2-1

Pheromone Reinforced Edges:

1-4, 4-3, 3-5, 5-2, 2-1

Pheromone deposits = $1/\text{Length}$
= $1/52 = 0.0192$

Then ant 2 will provide additional pheromone for $1/60 = 0.0167$ in segments traversed (Ant 2: 1-4- 2-5-3-1), as well as reduced the evaporation rate $(1-0.5) \times 1 = 0.5$. So will we get the pheromone matrix

0.5000	0.5000	0.5000	0.5359	0.5000
0.5192	0.5000	0.5000	0.5000	0.5167
0.5167	0.5000	0.5000	0.5000	0.5192
0.5000	0.5167	0.5192	0.5000	0.5000
0.5000	0.5192	0.5167	0.5000	0.5000

Fig. 7: Pheromone Matrix after Ant 1 and Ant 2 update their pheromone trail

Tour = 1-4-2-5-3-1

Pheromone Reinforced Edges:

1-4, 4-2, 2-5, 5-3, 3-1

Pheromone deposits = $1/\text{Length}$
= $1/60 = 0.0167$

Ant 3 also add an additional pheromone $1/60 = 0.0167$. So that for the segments traversed (Ant 3: 1 - 4-5 - 2 - 3-1), the pheromone matrix becomes

0.5000	0.5000	0.5000	0.5526	0.5000
0.5192	0.5000	0.5167	0.5000	0.5167
0.5334	0.5000	0.5000	0.5000	0.5192
0.5000	0.5167	0.5192	0.5000	0.5167
0.5000	0.5359	0.5167	0.5000	0.5000

Fig. 8: Pheromone Matrix after All 3 Ants update their Pheromone trails

Tour = 1-4-5-2-3-1

Pheromone Reinforced Edges:

1-4, 4-5, 5-2, 2-3, 3-1

Pheromone deposits = $1/\text{Length}$
= $1/60 = 0.0167$

Edge 1-4 gets Pheromone boost as many as 3 times, because it is included in all three paths/tours

Then in the next iteration we repeat from step 2 until the maximum number of iterations is achieved.

Example

Running the code in Matlab.

```
>> d=[0 10 12 11 14
```

```
10 0 13 15 8
```

```
12 13 0 9 14
```

```
11 15 9 0 16
```

```
14 8 14 16 0]
```

```
>> [bestroute,mincost]=acs_budi(d,20,10)
```

```
bestroute =
```

```
1 4 3 5 2 1
```

```
mincost =
```

```
52
```

Matlab Code

```
function [besttour,mincost]=acs_budi(d,iter,n_ants);
%iter=100;%number of cycles.
% d is n x n distance matrix
%xy is coordinates of cities m x 2 (n is number of cities)
m=n_ants;%200;%number of ants.
n=length(d);%jumlah kota.
e=.5;%evaporation coefficient.
```

```

alpha=1;%order of effect of ants' sight.
beta=2;%pangkat order of trace's effect.
%for i=1:n
    %   for j=1:n
        %   d(i,j)=sqrt((xy(i,1)-xy(j,1))^2+(xy(i,2)-xy(j,2))^2);
            %menghitung jarak antar kota
        %   end
    %end

for i=1:n%generating sight matrix.
    for j=1:n
        if d(i,j)==0
            h(i,j)=0;
        else
            h(i,j)=1/d(i,j);%inverse distance
        end
    end
end

thoinit=.1*ones(n);%initial tho.
%el=.96;%coefficient of common cost elimination.
for i=1:iter
    for i=1:m
        %ants initial placing.
        rute(i,1)=1;%ants start from city 1
    end
    %ant tour
    for i=1:m %for all ants
        mh=h;%invers distance matrix
        for j=1:n%next node
            c=rute(i,j);%choose next city to visit
            mh(:,c)=0;%for column c,inv distance=0
            tho=(thoinit(c,:).^beta).*(mh(c,:).^alpha);%update pheromone
            s=(sum(tho));%jumlah tho
            p=(1/s).*tho;%probability
        %
            p
            r=rand;
            s=0;
            for k=1:n %number of city
                s=s+p(k);%use roulette wheel
                if r<=s
                    rute(i,j+1)=k;%penempatan semut i di simpul berikutnya
                    break
                end
            end
        end
    end
end

%%%% resulting tour/route
rute_c=horzcat(rute,rute(:,1));%add first city to the tour/route
% %compute total distance of path traverse by each ant
for i=1:m %number of ant

```

```

s=0;
for j=1:n-1
    s=s+d(rute_c(i,j),rute_c(i,j+1));
end

f(i)=s;
end
jaraktot=f;
[minf,idk]=min(f);
ter=rute_c(idk,:);
%f=f-el*min(f);%elimination of common cost.
%update edge belonging to Lbest

thoinit=(1-e)*thoinit;
for i=1:m % for all ant
    % for i=idk;
    for j=1:n-1 % for all cities
        dt=1/f(i);%inverse total distance, delta pheromone
        %update pheromone

thoinit(rute_c(i,j),rute_c(i,j+1))=thoinit(rute_c(i,j),rute_c(i,j+1))+
dt;%updating traces.
        % thoinit(rute_c(i,j),rute_c(i,j+1))=(1-
e)*thoinit(rute_c(i,j),rute_c(i,j+1))+dt;%updating traces.

    end
end
%thoinit;
% costoa(i)=mean(cost);
%[mincost(i),number]=min(jaraktot);
%besttour(i,:)=rute_c(number,:);
%iteration(i)=i;
end
besttour=ter;
mincost=jartsp(besttour,d);

function jarak=jartsp(x1,dx)
%fungsi tsp
%input:
    %x1= rute tsp (contoh: 1 2 3 4 5 1, 1=kota asal, 2345=kota tujuan
    %dx= matrik jarak rute tsp
%output:
    %jarak = jarak total rute tsp

[r,c]=size(x1);
k=c-1;%jumlah kota dalam rute tsp
s=0; %jarak/posisi awal di kota pertama
for j=1:k
    s=s+dx(x1(j),x1(j+1)); %peng akumulasian jarak rute tsp
end
jarak=s;

```

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